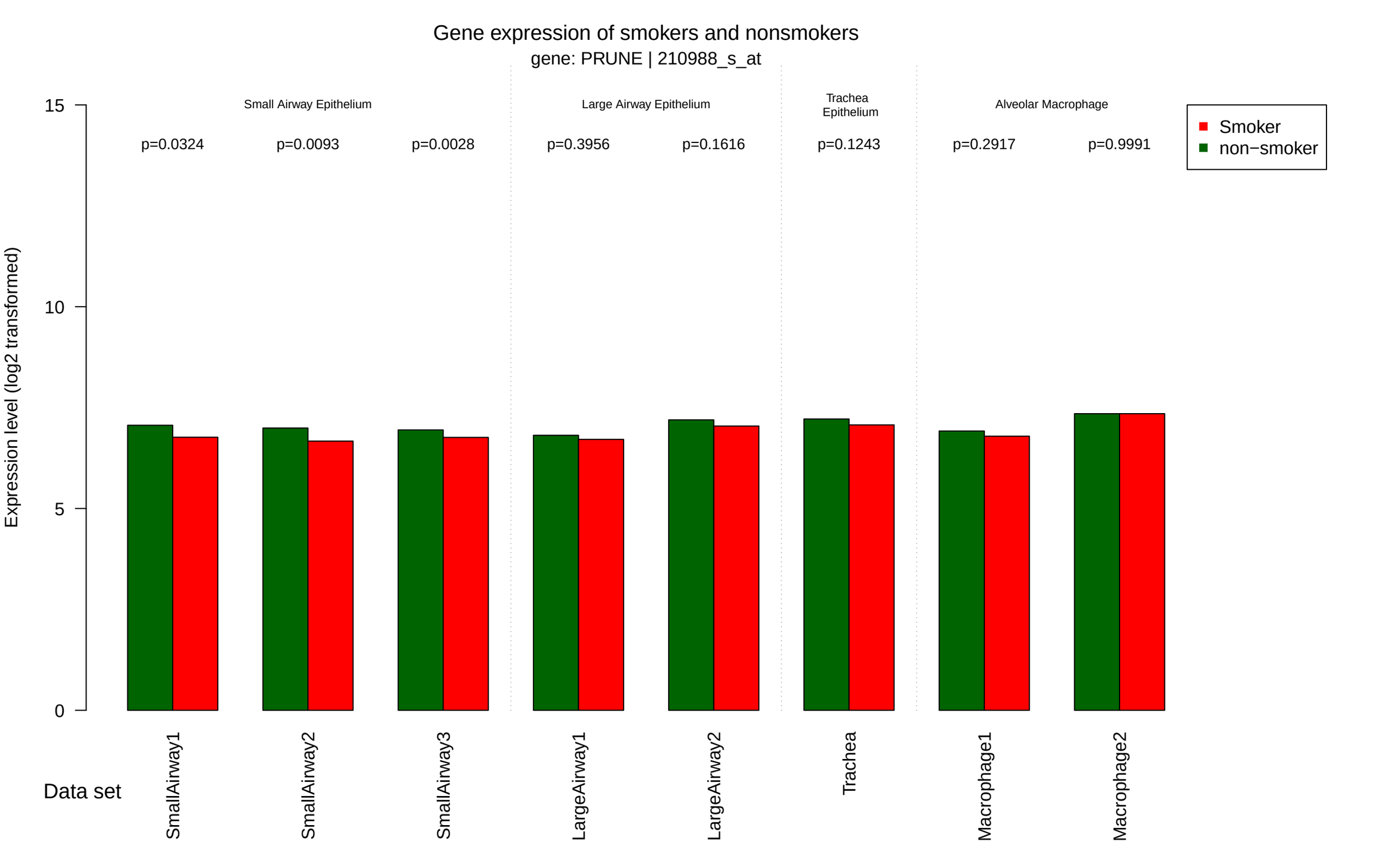
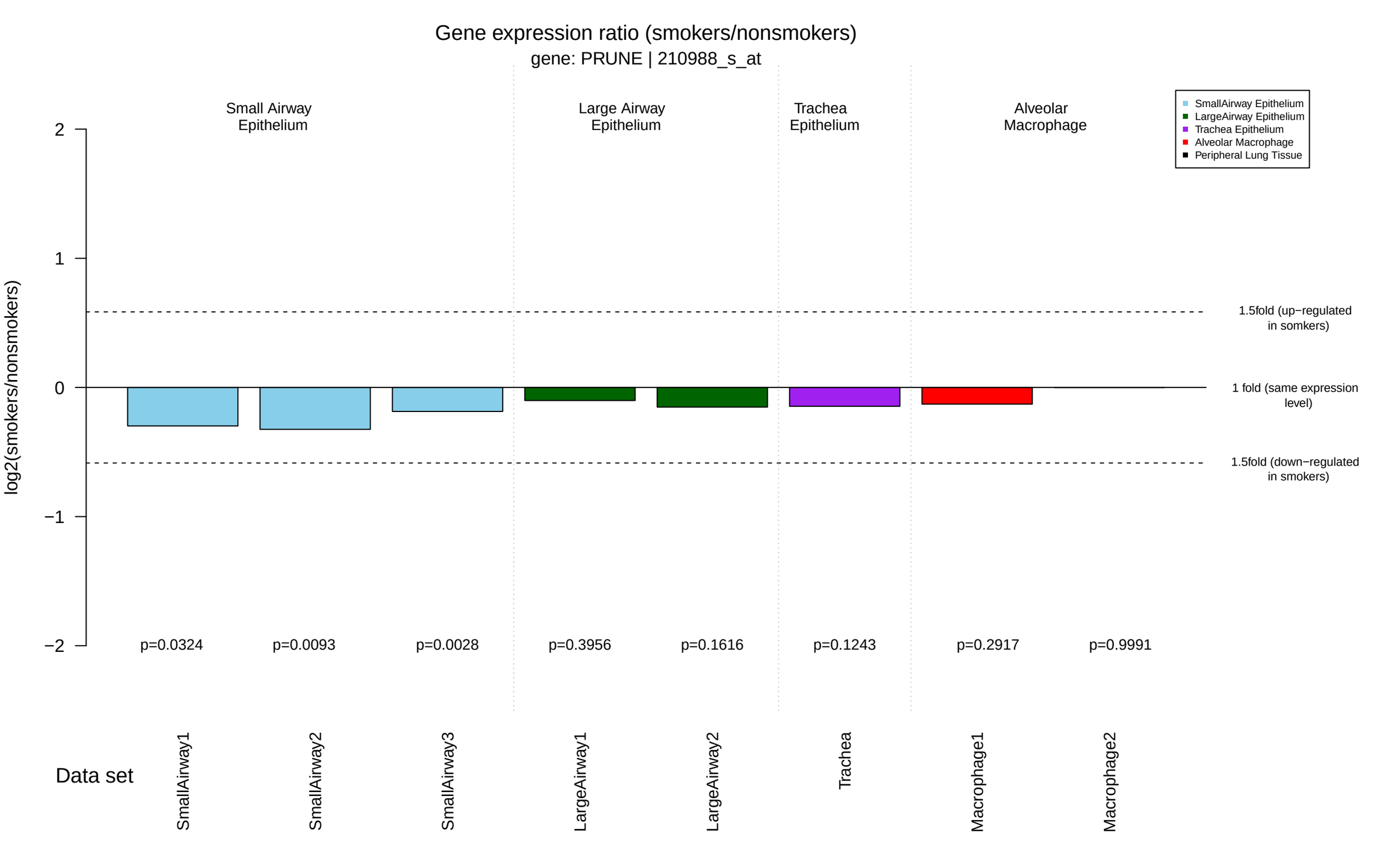
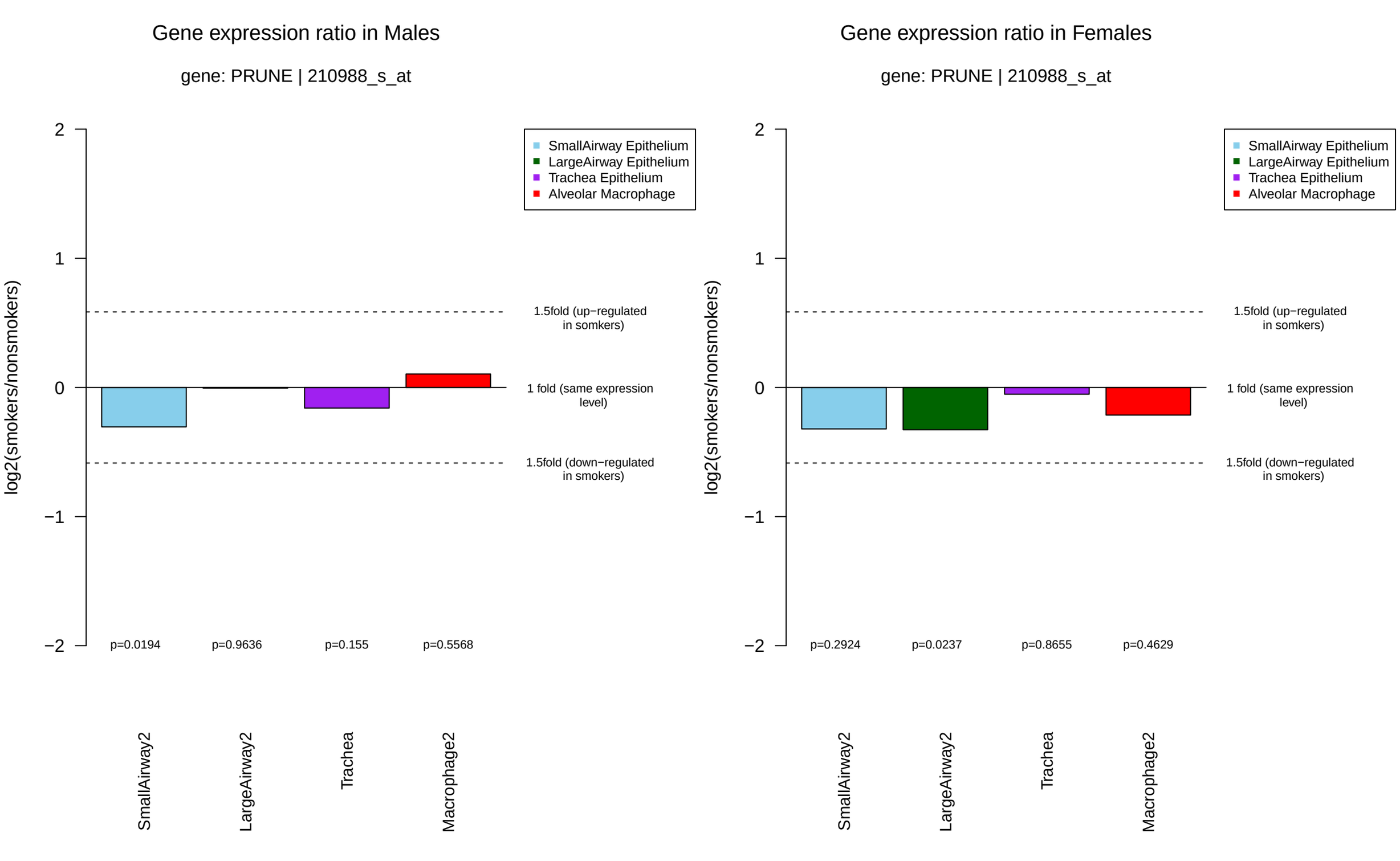
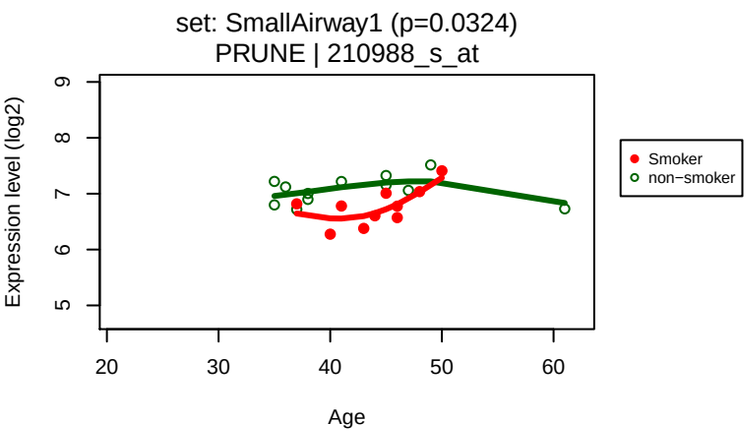
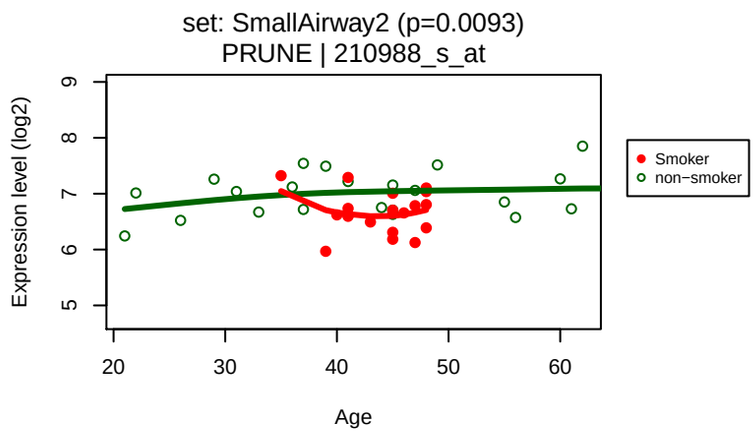
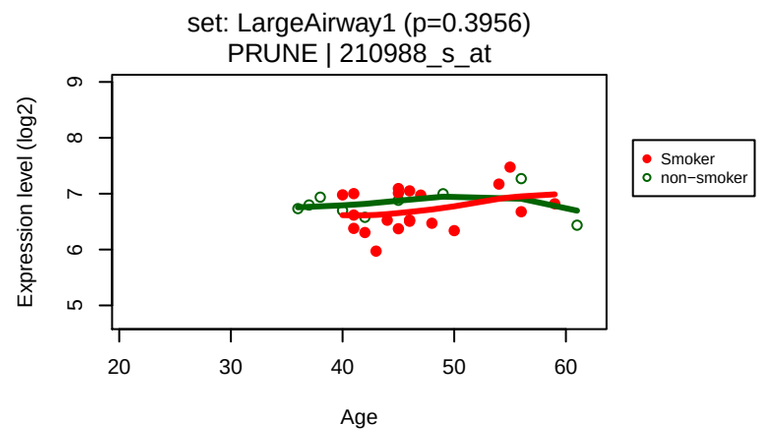

Gene expression of smokers and nonsmokers
gene: PRUNE | 210988_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0324
p=0.0093
p=0.0028
p=0.3956
p=0.1616
p=0.1243
p=0.2917
p=0.9991
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

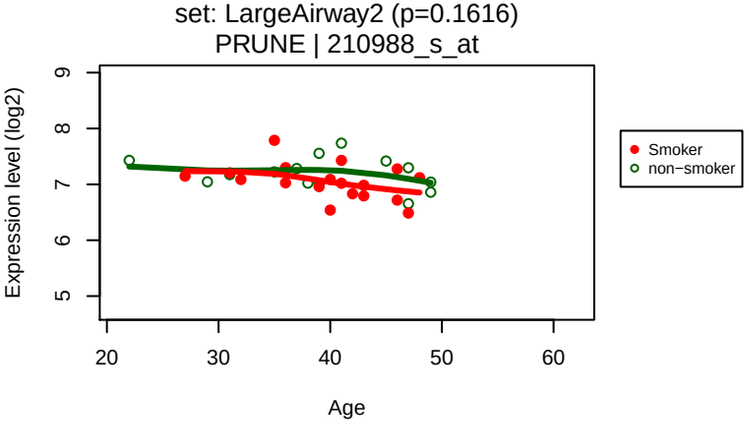

Gene expression ratio (smokers/nonsmokers)
gene: PRUNE | 210988_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0324
p=0.0093
p=0.0028
p=0.3956
p=0.1616
p=0.1243
p=0.2917
p=0.9991
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

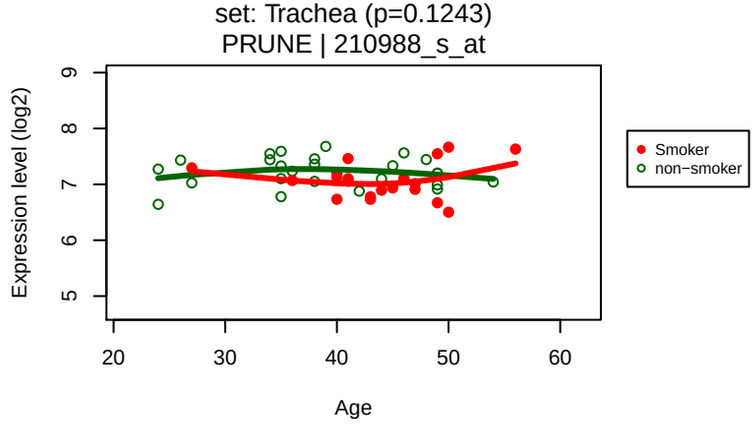

Gene expression ratio in Males
gene: PRUNE | 210988_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0194
p=0.9636
p=0.155
p=0.5568
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRUNE | 210988_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2924
p=0.0237
p=0.8655
p=0.4629
SmallAirway2
LargeAirway2
Trachea
Macrophage2

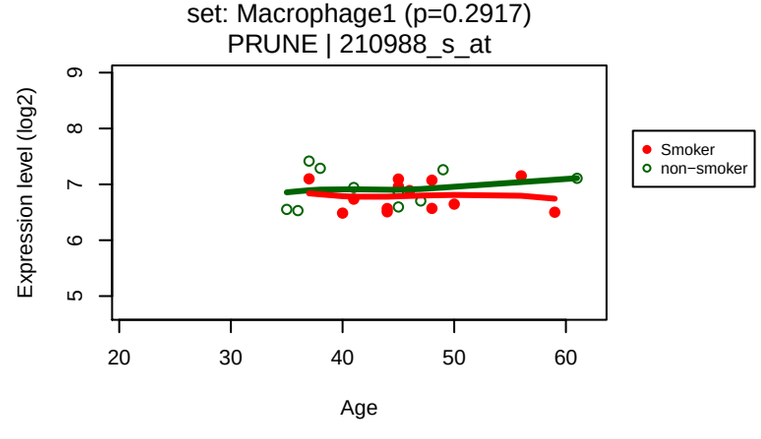

set: SmallAirway1 (p=0.0324)
PRUNE | 210988_s_at
Expression level (log2)
Age
Smoker
non−smoker

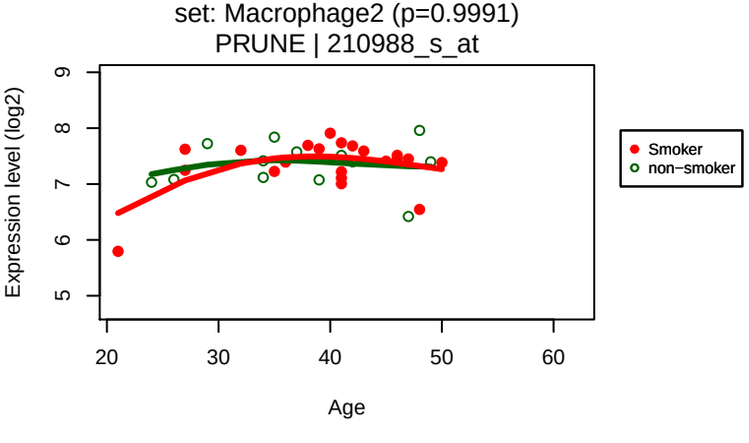

set: SmallAirway2 (p=0.0093)
PRUNE | 210988_s_at
Expression level (log2)
Age
Smoker
non−smoker

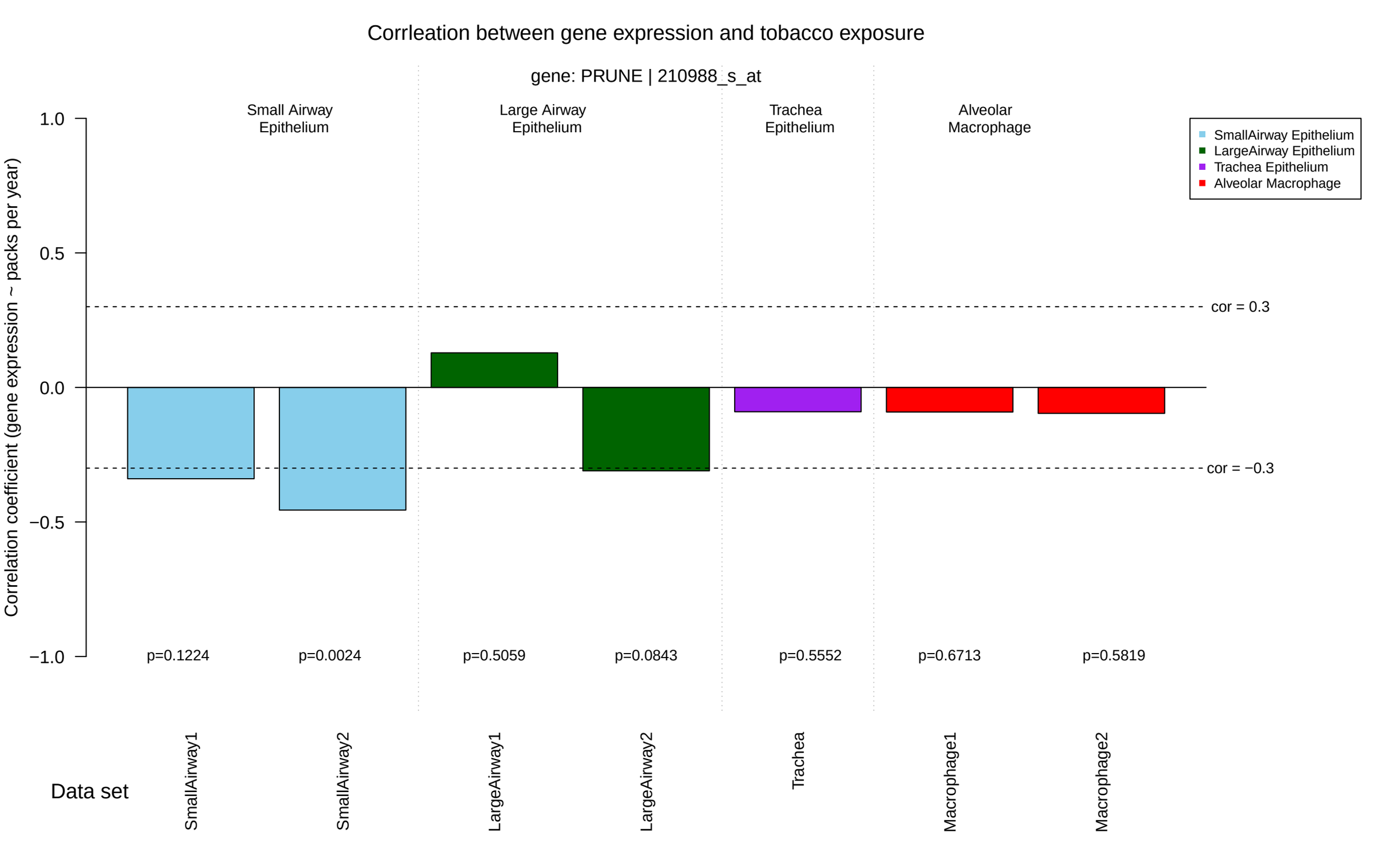

set: LargeAirway1 (p=0.3956)
PRUNE | 210988_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1616)
PRUNE | 210988_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1243)
PRUNE | 210988_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2917)
PRUNE | 210988_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9991)
PRUNE | 210988_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRUNE | 210988_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1224
p=0.0024
p=0.5059
p=0.0843
p=0.5552
p=0.6713
p=0.5819
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2